# Kenexai Hackathon 2k26: Data & AI Challenges @ CHARUSAT, Changa (24hrs Round 2)

The problems in this hackathon are designed to reflect real-world challenges in the fields of Data Engineering and Artificial Intelligence/Machine Learning. Participants are expected to work with all or combination of following task in the step wise for achieving the expected outcome of the problem definition.

**Data & AI Challenge Road Map**

1. Download the dataset from Kaggle/similar or synthetic data generation
2. Data source simulation (Generating data at a given interval)
3. Data warehouse design (using best practices like dimensionality modelling, star/snowflake schema, medallion architecture, data vault design),
4. ETL pipeline implementation (data preprocessing steps like handling missing values, normalization, outlier analysis, value and type conversion, data profiling, quality checks)
5. Provide a dashboard to review data quality and EDA.
6. Identify personas and key KPIs for each persona and build an analytical dashboard for each persona.
7. Identify ML use cases (i.e., regression, classification, clustering, association, etc) and implement the same.
8. Identify GenAI use cases (i.e., conversational RAG chatbot, document/data query using NLP, extract useful information from unstructured documents like pdfs, docs, webpages, etc.)
9. Identify optimization of use cases.
10. Dockerized deployment of end-to-end application

**Dataset Link:** https://github.com/Ridgeant/Kenexai-Hackathon-2k26-Charusat-R2

| **KD&A-1** |
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| **Title: Patient Readmission Prediction System**<br>Hospital readmissions within 30 days cost healthcare systems billions annually and often indicate gaps in treatment quality. This problem asks students to build a predictive system using real Electronic Health Record (EHR) data. The goal is to identify high-risk patients BEFORE they are discharged so doctors can intervene early. Students will work with clinical data including diagnoses (ICD codes), lab results, medication history, demographics, and prior admission records. |
| **KD&A-2**<br><br>**Title: E-Commerce Customer Churn Prediction & Retention** |

E-commerce platforms lose 20–30% of customers annually to churn. Acquiring a new customer costs 5x more than retaining one. This problem asks students to predict which customers are likely to churn (stop purchasing) and then use AI to create personalized retention messages for each at-risk customer. Students will work with customer transaction history, browsing behavior, complaint logs, and demographic data to build both a predictive model and a GenAI-powered communication system.

## KD&A-3

**Title: Athlete Performance Analytics & Injury Risk Prediction**

Sports teams spend millions on player acquisition, but many investments fail due to unexpected injuries or performance decline. Data analytics can identify underperforming players early and flag injury risk before it happens. This project asks students to analyze historical match statistics and wearable sensor data to build a player performance scoring system, predict injury risk, and use GenAI to generate personalized coaching recommendations for each player — mimicking the real tools used by clubs like Manchester City, FC Barcelona, and IPL teams.

## KD&A-4

**Title: Alert Incident Intelligence**

Modern IT systems generate thousands of alerts from monitoring tools such as servers, applications, networks, and security platforms. Many of these alerts are duplicated, noisy, or unrelated, making it difficult for Site Reliability Engineering and operations teams to quickly identify the real root cause of incidents. This project focuses on building an dashboard of Incident Intelligence platform that ingests alerts from multiple monitoring tools, normalizes and deduplicates them, and correlates related alerts into meaningful incidents. Students will design a data pipeline to process alerts and apply analytics to detect patterns in recurring failures. Additionally, a GenAI-powered chatbot or AI Copilot will be developed to automatically generate incident summaries, suggest possible root causes, and recommend resolution steps. The system will help teams investigate incidents faster, reduce alert fatigue, and improve overall system reliability.

## KD&A-5

**Title: Intern Activity Intelligence & Productivity Analytics Platform**

Organizations often collect daily activity reports from interns or trainees through collaboration platforms like Microsoft Teams or internal reporting systems. These reports usually contain information such as tasks performed, technologies used (e.g., Python coding, SQL, PySpark, NumPy, Pandas, data engineering tasks), time spent on different activities, project contributions, and progress recorded in

learning management systems. However, this data is often unstructured and difficult for managers to analyze at a scale.

## KD&A-6

**Title: Resume Screening & Talent Analytics Platform**

Recruitment teams receive a large number of resumes for every job opening, making manual screening time-consuming and inefficient. Many qualified candidates may be overlooked due to human bias or lack of automated tools. This project focuses on building a Resume Screening & Talent Analytics Platform that automatically processes resumes and matches them with job descriptions using data analytics and AI techniques. Missing PII data should be synthesized. Resume data should also be augmented.

## KD&A-7

**Title: AI-Powered Customer Support Call Quality Analyzer**

Customer support teams receive thousands of calls every day, and while these conversations contain valuable insights about customer problems, service quality, and agent performance, it is practically impossible for managers to manually review and understand what is happening across all calls. As a result, organizations often have very limited visibility into customer concerns, call outcomes, and overall support quality, even though they possess large volumes of call transcripts and related metadata (such as call ID, agent ID, timestamps, and duration). The challenge is to design an AI-driven system that can analyze these call transcripts at scale and help the business extract meaningful insights from them. Additionally, the organization evaluates calls based on a hierarchical quality framework consisting of Sections → Sub-sections → Criteria, each having configurable weights. This evaluation structure can evolve over time, meaning business users should be able to add, remove, modify, enable, or disable criteria and adjust their weights without requiring code changes, and the system should adapt accordingly. The goal is to build a scalable, configurable solution that can process large volumes of call conversations, evaluate them using the dynamic framework, and provide better visibility into what is happening across customer support interactions.

## KD&A-8

**Title: Vehicle Insurance Risk Analytics & Claim Prediction Platform**

Car insurance companies manage large volumes of policyholder and vehicle data, including information such as vehicle type, production year, insured value, premium amount, and claim history. However,

insurers often struggle to accurately assess risk levels and predict potential claims using traditional manual analysis methods. This can lead to incorrect premium pricing, higher claim losses, and inefficient risk management. This project focuses on building a Vehicle Insurance Risk Analytics & Claim Prediction Platform that analyzes insurance policy data such as policy duration, vehicle specifications (make, seats, engine capacity), insured value, premium amount, vehicle usage, and previous claim payments. Students will apply data analytics and machine learning techniques to identify patterns that influence insurance claims and risk levels. Data synthesize and data stiching is required.

**KD&A-9**

**Title: Student Academic Success, Subject Performance & Career Readiness Analytics Platform**

Educational institutions collect large amounts of student data including exam results, subject marks, prior academic records, lifestyle habits, and career preferences. However, these datasets are often stored in separate systems and are rarely analyzed together. As a result, institutions miss valuable insights about subject difficulty, learning gaps, and how factors such as study habits, stress levels, financial conditions, or part-time work influence academic performance and career readiness. This project focuses on building a Student Academic Success & Career Readiness Analytics Platform by integrating multiple datasets through data stitching. The platform will analyze subject-wise marks, SGPA/CGPA trends, backlog history, study behavior, and lifestyle factors to identify learning gaps, detect students at risk, and predict academic performance trends. Using data analytics and machine learning, the system will generate insights for faculty, mentors, and academic administrators, helping them improve teaching strategies, provide early academic support, and enhance career guidance for students. Use if data stitching require.

**KD&A-10**

**Title: ICC Men's T20 Cricket Match World Cup 2026 Outcome Prediction**

Cricket teams increasingly rely on data analytics to optimize match strategies, player selection, and performance evaluation. Modern cricket datasets contain detailed information such as player statistics, match outcomes, venue conditions, ball-by-ball events, and team compositions. By analyzing these large datasets, teams and analysts can identify patterns that influence match outcomes and improve decision-making.